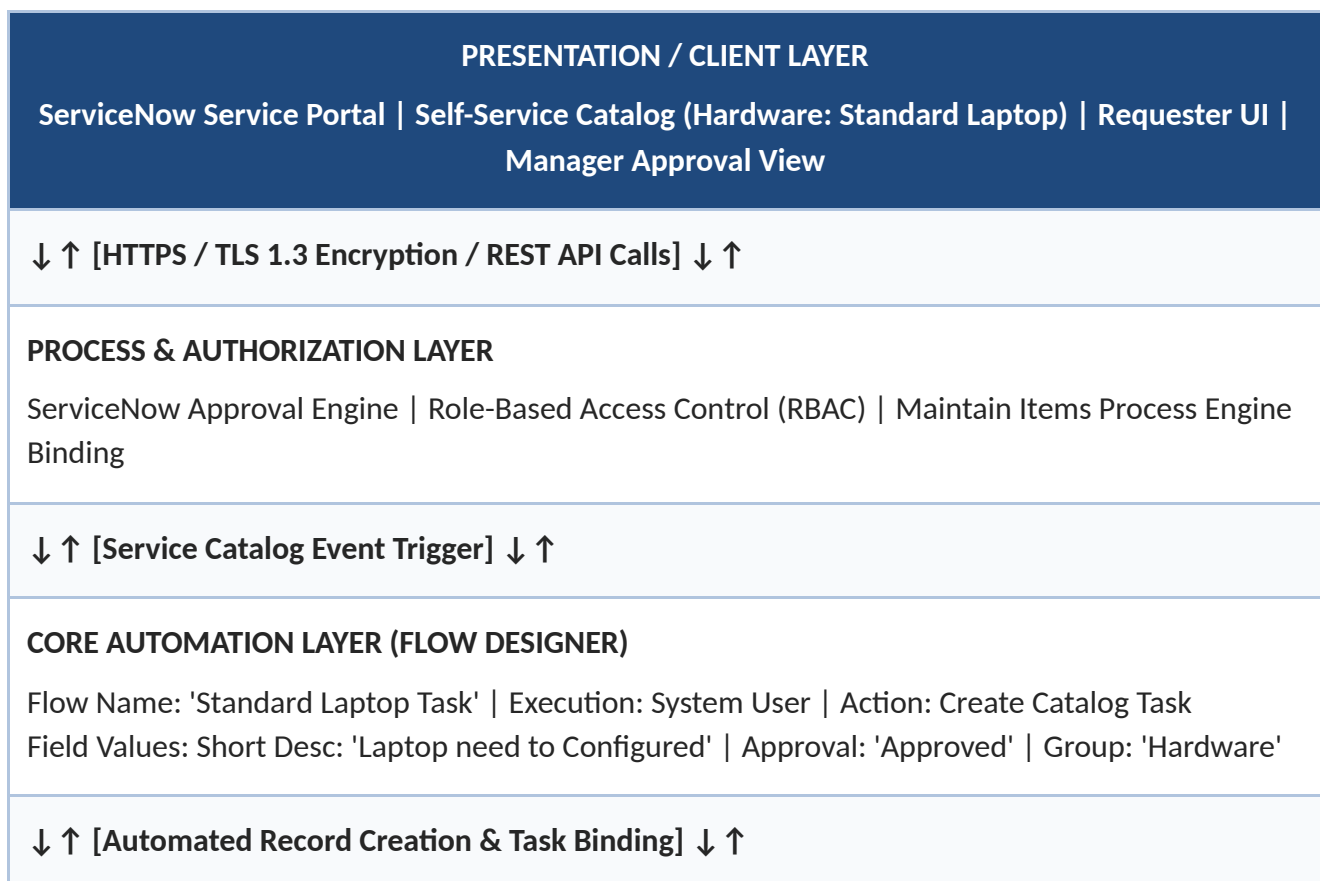


Solution Architecture

Date	23/09/2026
Team ID	SWTID-2026-3089
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Institution	Arignar Anna College Code: MSU 302
Team Members	Aldhaf F (Leader), Bhuvanasri S, Dhanisha Muhi N S, Divya Lakshmi K, Dharshini P
Maximum Marks	5 Marks

Solution Architecture Diagram

Provide a high-level visual representation of the solution's multi-tier architecture. The diagram maps the Presentation Layer, Process & Automation Layer, Core Workflow Services, and Cloud Data Storage Layer:



CLOUD DATA & STORAGE LAYER

ServiceNow Relational Datastore: sc_request, sc_req_item, sc_task, sys_user, sys_user_group
(Hardware Team)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Employee self-service catalog interface for ordering standard laptops, selecting hardware options, and viewing request approval status.	ServiceNow Service Portal, Service Catalog, HTML5, CSS3, Angular Widgets
Process & Approval Engine	Manages managerial approval records, validates order criteria, and communicates approval state changes to initiate downstream automation.	ServiceNow Approval Engine, ITIL Process Workflows, SysApproval Tables
Flow Designer Orchestrator	Event-driven visual automation engine that executes under elevated System User authority, listens to Service Catalog triggers, and creates Catalog Tasks.	ServiceNow Flow Designer, Process Automation Suite, Global Scope
Hardware Task Management	Dedicated task subsystem assigning tasks directly to Hardware group technicians with standardized configuration instructions ('Laptop need to Configured').	ServiceNow Catalog Task ('sc_task'), Assignment Group Routing
Data & Cloud Storage Layer	Relational cloud storage layer maintaining complete relational links between REQ, RITM, SCTASK, user profiles, and hardware inventory records.	ServiceNow Enterprise Cloud Database (Relational Schema, MariaDB/MySQL)